

Rapid Object Detection using a Boosted Cascade of Simple Features

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Abstract

This paper describes a machine learning approach for visual object detection which is capable of processing images extremely rapidly and achieving high detection rates. This work is distinguished by three key contributions. The first is the introduction of a new image representation called the “Integral Image” which allows the features used by our detector to be computed very quickly. The second is a learning algorithm, based on AdaBoost, which selects a small number of critical visual features from a larger set and yields extremely efficient classifiers[6]. The third contribution is a method for combining increasingly more complex classifiers in a “cascade” which allows background regions of the image to be quickly discarded while spending more computation on promising object-like regions. The cascade can be viewed as an object specific focus-of-attention mechanism which unlike previous approaches provides statistical guarantees that discarded regions are unlikely to contain the object of interest. In the domain of face detection the system yields detection rates comparable to the best previous systems. Used in real-time applications, the detector runs at 15 frames per second without resorting to image differencing or skin color detection.

1. Introduction

This paper brings together new algorithms and insights to construct a framework for robust and extremely rapid object detection. This framework is demonstrated on, and in part motivated by, the task of face detection. Toward this end we have constructed a frontal face detection system which achieves detection and false positive rates which are equivalent to the best published results [16, 12, 15, 11, 1]. This face detection system is most clearly distinguished from previous approaches in its ability to detect faces extremely rapidly. Operating on 384 by 288 pixel images, faces are de-

tected at 15 frames per second on a conventional 700 MHz Intel Pentium III. In other face detection systems, auxiliary information, such as image differences in video sequences, or pixel color in color images, have been used to achieve high frame rates. Our system achieves high frame rates working only with the information present in a single grey scale image. These alternative sources of information can also be integrated with our system to achieve even higher frame rates.

There are three main contributions of our object detection framework. We will introduce each of these ideas briefly below and then describe them in detail in subsequent sections.

The first contribution of this paper is a new image representation called an *integral image* that allows for very fast feature evaluation. Motivated in part by the work of Papa-georgiou et al. our detection system does not work directly with image intensities [10]. Like these authors we use a set of features which are reminiscent of Haar Basis functions (though we will also use related filters which are more complex than Haar filters). In order to compute these features very rapidly at many scales we introduce the integral image representation for images. The integral image can be computed from an image using a few operations per pixel. Once computed, any one of these Harr-like features can be computed at any scale or location in *constant* time.

The second contribution of this paper is a method for constructing a classifier by selecting a small number of important features using AdaBoost [6]. Within any image sub-window the total number of Harr-like features is very large, far larger than the number of pixels. In order to ensure fast classification, the learning process must exclude a large majority of the available features, and focus on a small set of critical features. Motivated by the work of Tieu and Viola, feature selection is achieved through a simple modification of the AdaBoost procedure: the weak learner is constrained so that each weak classifier returned can depend on only a

使用简单特征的增强级联进行快速目标检测

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摘要

This paper describes a machine learning approach for visual object detection which is capable of processing images extremely rapidly and achieving high detection rates. This work is distinguished by three key contributions. The first is the introduction of a new image representation called the “Integral Image” which allows the features used by our detector to be computed very quickly. The second is a learning algorithm, based on AdaBoost, which selects a small number of critical visual features from a larger set and yields extremely efficient classifiers[6]. The third contribution is a method for combining increasingly more complex classifiers in a “cascade” which allows background regions of the image to be quickly discarded while spending more computation on promising object-like regions. The cascade can be viewed as an object specific focus-of-attention mechanism which unlike previous approaches provides statistical guarantees that discarded regions are unlikely to contain the object of interest. In the domain of face detection the system yields detection rates comparable to the best previous systems. Used in real-time applications, the detector runs at 15 frames per second without resorting to image differencing or skin color detection.

1. 引言

本文汇集了新的算法与洞见，构建了一个鲁棒且极速的目标检测框架。该框架以人脸检测任务为演示对象，并部分受其启发而构建。为此，我们开发了一个正面人脸检测系统，其检测率和误报率均与已发表的最佳成果相当[16, 12, 15, 11, 1]。该人脸检测系统最显著的特点在于其极快的检测速度：在384×288像素的图像上，人脸检测仅需——

在传统的700 MHz英特尔奔腾III处理器上，该系统以每秒15帧的速度进行检测。在其他面部检测系统中，通常需要借助辅助信息——例如视频序列中的图像差异，或彩色图像中的像素颜色——来实现高帧率。而我们的系统仅利用单张灰度图像中的信息，就能实现高帧率。若将这些其他信息源与我们的系统结合，还可实现更高的帧率。

我们的目标检测框架主要有三点贡献。下面我们将简要介绍这些想法，并在后续章节中详细阐述。

本文的第一个贡献是提出了一种称为integral image的新图像表示方法，该方法能够实现极快的特征评估。部分受到Papageorgiou等人研究的启发，我们的检测系统并不直接处理图像强度[10]。与这些研究者类似，我们采用了一组让人联想到哈尔基函数的特征集（同时我们也会使用比哈尔滤波器更复杂的相关滤波器）。为了在多尺度下快速计算这些特征，我们引入了图像的积分图表示法。积分图可通过每个像素的少量运算从图像中计算得出。一旦计算完成，任一此类类哈尔特征都可在constant时间内以任意尺度或位置进行计算。

本文的第二项贡献是提出了一种通过使用AdaBoost[6]选择少量重要特征来构建分类器的方法。在任何图像子窗口中，类Haar特征的总数非常庞大，远超像素数量。为了确保快速分类，学习过程必须排除大部分可用特征，并集中关注一小部分关键特征。受Tieu和Viola研究的启发，我们通过对AdaBoost流程进行简单修改来实现特征选择：弱学习器被约束为每个返回的弱分类器仅能依赖于一个

single feature [2]. As a result each stage of the boosting process, which selects a new weak classifier, can be viewed as a feature selection process. AdaBoost provides an effective learning algorithm and strong bounds on generalization performance [13, 9, 10].

The third major contribution of this paper is a method for combining successively more complex classifiers in a cascade structure which dramatically increases the speed of the detector by focusing attention on promising regions of the image. The notion behind focus of attention approaches is that it is often possible to rapidly determine where in an image an object might occur [17, 8, 1]. More complex processing is reserved only for these promising regions. The key measure of such an approach is the “false negative” rate of the attentional process. It must be the case that all, or almost all, object instances are selected by the attentional filter.

We will describe a process for training an extremely simple and efficient classifier which can be used as a “supervised” focus of attention operator. The term supervised refers to the fact that the attentional operator is trained to detect examples of a particular class. In the domain of face detection it is possible to achieve fewer than 1% false negatives and 40% false positives using a classifier constructed from two Harr-like features. The effect of this filter is to reduce by over one half the number of locations where the final detector must be evaluated.

Those sub-windows which are not rejected by the initial classifier are processed by a sequence of classifiers, each slightly more complex than the last. If any classifier rejects the sub-window, no further processing is performed. The structure of the cascaded detection process is essentially that of a degenerate decision tree, and as such is related to the work of Geman and colleagues [1, 4].

An extremely fast face detector will have broad practical applications. These include user interfaces, image databases, and teleconferencing. In applications where rapid frame-rates are not necessary, our system will allow for significant additional post-processing and analysis. In addition our system can be implemented on a wide range of small low power devices, including hand-helds and embedded processors. In our lab we have implemented this face detector on the Compaq iPaq handheld and have achieved detection at two frames per second (this device has a low power 200 mips *Strong Arm* processor which lacks floating point hardware).

The remainder of the paper describes our contributions and a number of experimental results, including a detailed description of our experimental methodology. Discussion of closely related work takes place at the end of each section.

2. Features

Our object detection procedure classifies images based on the value of simple features. There are many motivations

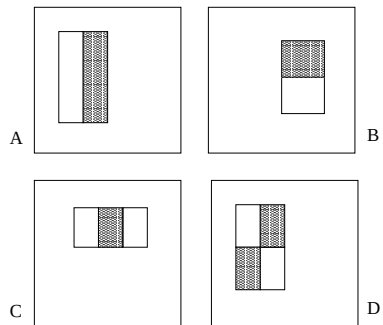


Figure 1: Example rectangle features shown relative to the enclosing detection window. The sum of the pixels which lie within the white rectangles are subtracted from the sum of pixels in the grey rectangles. Two-rectangle features are shown in (A) and (B). Figure (C) shows a three-rectangle feature, and (D) a four-rectangle feature.

for using features rather than the pixels directly. The most common reason is that features can act to encode ad-hoc domain knowledge that is difficult to learn using a finite quantity of training data. For this system there is also a second critical motivation for features: the feature based system operates much faster than a pixel-based system.

The simple features used are reminiscent of Haar basis functions which have been used by Papageorgiou et al. [10]. More specifically, we use three kinds of features. The value of a *two-rectangle feature* is the difference between the sum of the pixels within two rectangular regions. The regions have the same size and shape and are horizontally or vertically adjacent (see Figure 1). A *three-rectangle feature* computes the sum within two outside rectangles subtracted from the sum in a center rectangle. Finally a *four-rectangle feature* computes the difference between diagonal pairs of rectangles.

Given that the base resolution of the detector is 24x24, the exhaustive set of rectangle features is quite large, over 180,000. Note that unlike the Haar basis, the set of rectangle features is overcomplete¹.

2.1. Integral Image

Rectangle features can be computed very rapidly using an intermediate representation for the image which we call the integral image.² The integral image at location x, y contains the sum of the pixels above and to the left of x, y , inclusive:

$$ii(x, y) = \sum_{x' \leq x, y' \leq y} i(x', y'),$$

¹A complete basis has no linear dependence between basis elements and has the same number of elements as the image space, in this case 576. The full set of 180,000 thousand features is many times over-complete.

²There is a close relation to “summed area tables” as used in graphics [3]. We choose a different name here in order to emphasize its use for the analysis of images, rather than for texture mapping.

单一特征[2]。因此，提升过程的每个阶段（即选择新弱分类器的过程）可视为一种特征选择过程。AdaBoost 提供了一种高效的学习算法，并在泛化性能上给出了强有力的理论保证[13, 9, 10]。

本文的第三个主要贡献是提出了一种在级联结构中组合逐渐复杂分类器的方法，该方法通过将注意力集中在图像中有前景的区域，显著提高了检测器的速度。注意力聚焦方法背后的理念是，通常能够快速确定物体可能出现在图像中的位置[17, 8, 1]。更复杂的处理仅保留给这些有前景的区域。这种方法的关键衡量标准是注意力过程的“漏检率”。必须确保所有或几乎所有的物体实例都被注意力过滤器选中。

我们将描述一种训练极其简单高效分类器的过程，该分类器可用作“有监督”的注意力聚焦算子。所谓“有监督”，是指该注意力算子经过训练能够检测特定类别的样本。在人脸检测领域，使用由两个类哈尔特征构建的分类器，可以实现低于1%的漏检率和40%的误检率。该滤波器的作用是将最终检测器需要评估的位置数量减少一半以上。

那些未被初始分类器拒绝的子窗口会由一系列分类器处理，每个分类器都比前一个略微复杂。如果任何分类器拒绝了子窗口，则不再进行后续处理。这种级联检测过程的结构本质上是一个退化决策树，因此与Geman及其同事的研究[1, 4]相关。

一个极其快速的人脸检测器将拥有广泛的实际应用。这些应用包括用户界面、图像数据库和视频会议。在不需要高帧率的应用中，我们的系统将允许进行大量额外的后处理和分析。此外，我们的系统可以在各种小型低功耗设备上实现，包括手持设备和嵌入式处理器。在我们的实验室中，我们已经在康柏iPaq手持设备上实现了这个人脸检测器，并达到了每秒两帧的检测速度（该设备配备了一个低功耗的200 MIPS Strong Arm处理器，且缺乏浮点运算硬件）。

本文的其余部分描述了我们的贡献和一系列实验结果，包括对我们实验方法的详细说明。对密切相关工作的讨论将在每节末尾进行。

2. 特性

我们的目标检测程序基于简单特征值对图像进行分类。其动机有很多

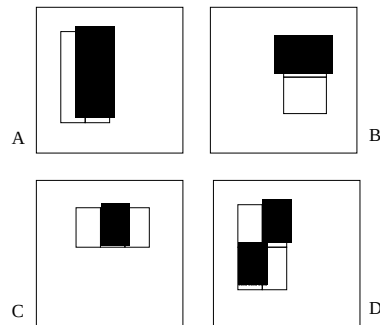


图1：相对于封闭检测窗口显示的示例矩形特征。灰色矩形内像素之和减去白色矩形内像素之和。(A)和(B)展示的是双矩形特征，(C)展示的是三矩形特征，(D)展示的是四矩形特征。

使用特征而非直接使用像素的最常见原因是，特征能够编码特定领域的先验知识，这些知识难以通过有限数量的训练数据来学习。对于该系统而言，采用特征还有第二个关键动机：基于特征的系统运行速度远快于基于像素的系统。

所使用的简单特征让人联想到Papageorgiou等人[10]曾使用的哈尔基函数。更具体地说，我们使用了三种特征。*two-rectangle feature*的值是两个矩形区域内像素总和的差值。这些区域具有相同的大小和形状，并且水平或垂直相邻（见图1）。*three-rectangle feature*计算中心矩形内像素总和减去两个外部矩形内像素总和的结果。最后，*four-rectangle feature*计算对角线矩形对之间的差值。

鉴于检测器的基础分辨率为24x24，矩形特征的穷举集相当庞大，超过18万个。请注意，与哈尔基不同，矩形特征集是过完备的¹。

2.1. 积分图像

矩形特征可以使用一种称为积分图像的中间表示来非常快速地计算。²位置 x_i, y_i 处的积分图像包含 x_i, y_i 上方和左侧（包括 x_i, y_i 本身）的像素之和：

$$I(x_i, y_i) = \sum_{x_1 \leq x_i, y_1 \leq y_i} I(x_1, y_1)$$

一个完整的基在基元素之间没有线性依赖，并且与图像空间具有相同数量的元素，在本例中为576。完整的18万个特征集是多次过完备的。

²这与图形学中使用的“积分面积表”[3]有密切关系。我们在此选择不同的名称，是为了强调其在图像分析中的应用，而非纹理映射。

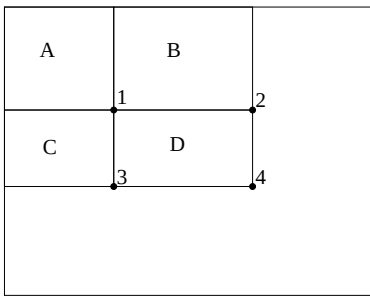


Figure 2: The sum of the pixels within rectangle D can be computed with four array references. The value of the integral image at location 1 is the sum of the pixels in rectangle A . The value at location 2 is $A + B$, at location 3 is $A + C$, and at location 4 is $A + B + C + D$. The sum within D can be computed as $4 + 1 - (2 + 3)$.

where $ii(x, y)$ is the integral image and $i(x, y)$ is the original image. Using the following pair of recurrences:

$$s(x, y) = s(x, y - 1) + i(x, y) \quad (1)$$

$$ii(x, y) = ii(x - 1, y) + s(x, y) \quad (2)$$

(where $s(x, y)$ is the cumulative row sum, $s(x, -1) = 0$, and $ii(-1, y) = 0$) the integral image can be computed in one pass over the original image.

Using the integral image any rectangular sum can be computed in four array references (see Figure 2). Clearly the difference between two rectangular sums can be computed in eight references. Since the two-rectangle features defined above involve adjacent rectangular sums they can be computed in six array references, eight in the case of the three-rectangle features, and nine for four-rectangle features.

2.2. Feature Discussion

Rectangle features are somewhat primitive when compared with alternatives such as steerable filters [5, 7]. Steerable filters, and their relatives, are excellent for the detailed analysis of boundaries, image compression, and texture analysis. In contrast rectangle features, while sensitive to the presence of edges, bars, and other simple image structure, are quite coarse. Unlike steerable filters the only orientations available are vertical, horizontal, and diagonal. The set of rectangle features do however provide a rich image representation which supports effective learning. In conjunction with the integral image, the efficiency of the rectangle feature set provides ample compensation for their limited flexibility.

3. Learning Classification Functions

Given a feature set and a training set of positive and negative images, any number of machine learning approaches

could be used to learn a classification function. In our system a variant of AdaBoost is used *both* to select a small set of features *and* train the classifier [6]. In its original form, the AdaBoost learning algorithm is used to boost the classification performance of a simple (sometimes called weak) learning algorithm. There are a number of formal guarantees provided by the AdaBoost learning procedure. Freund and Schapire proved that the training error of the strong classifier approaches zero exponentially in the number of rounds. More importantly a number of results were later proved about generalization performance [14]. The key insight is that generalization performance is related to the margin of the examples, and that AdaBoost achieves large margins rapidly.

Recall that there are over 180,000 rectangle features associated with each image sub-window, a number far larger than the number of pixels. Even though each feature can be computed very efficiently, computing the complete set is prohibitively expensive. Our hypothesis, which is borne out by experiment, is that a very small number of these features can be combined to form an effective classifier. The main challenge is to find these features.

In support of this goal, the weak learning algorithm is designed to select the single rectangle feature which best separates the positive and negative examples (this is similar to the approach of [2] in the domain of image database retrieval). For each feature, the weak learner determines the optimal threshold classification function, such that the minimum number of examples are misclassified. A weak classifier $h_j(x)$ thus consists of a feature f_j , a threshold θ_j and a parity p_j indicating the direction of the inequality sign:

$$h_j(x) = \begin{cases} 1 & \text{if } p_j f_j(x) < p_j \theta_j \\ 0 & \text{otherwise} \end{cases}$$

Here x is a 24x24 pixel sub-window of an image. See Table 1 for a summary of the boosting process.

In practice no single feature can perform the classification task with low error. Features which are selected in early rounds of the boosting process had error rates between 0.1 and 0.3. Features selected in later rounds, as the task becomes more difficult, yield error rates between 0.4 and 0.5.

3.1. Learning Discussion

Many general feature selection procedures have been proposed (see chapter 8 of [18] for a review). Our final application demanded a very aggressive approach which would discard the vast majority of features. For a similar recognition problem Papageorgiou et al. proposed a scheme for feature selection based on feature variance [10]. They demonstrated good results selecting 37 features out of a total 1734 features.

Roth et al. propose a feature selection process based on the Winnow exponential perceptron learning rule [11]. The Winnow learning process converges to a solution where many of these weights are zero. Nevertheless a very large

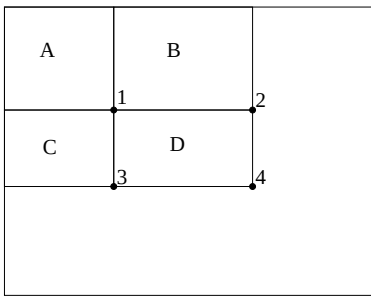


图2: 矩形内的像素和可通过四次数组引用计算得出。积分图像在位置1处的值为矩形A的像素和。位置2处的值为A+B, 位置3处的值为A+B+C, 位置4处的值为A+B+C+D。区域D内的像素和可通过 $s(4) - (s(3) + s(2) - s(1))$ 计算得出。

其中 $s(x, y)$ 是积分图像, $I(x, y)$ 是原始图像。使用以下递推对:

$$s(x, y) = s(x, y-1) + I(x, y) \quad (1)$$

$$s(x, y) = s(x-1, y) + s(x, y-1) - s(x-1, y-1) + I(x, y) \quad (2)$$

(其中 $s(x, 0)$ 是累积行和, $s(0, y) = 0$ 和 $s(-1, y) = 0$) 积分图像可在原始图像上单次遍历计算得出。

使用积分图像, 任何矩形和都可以通过四次数组引用计算得出(见图2)。显然, 两个矩形和之间的差异可以通过八次引用来计算。由于上述定义的双矩形特征涉及相邻的矩形和, 它们可以通过六次数组引用计算得出, 三矩形特征需要八次, 而四矩形特征则需要九次。

2.2. 特征讨论

矩形特征相较于可操纵滤波器[5,7]等替代方案而言较为原始。可操纵滤波器及其衍生方法在边界细节分析、图像压缩和纹理分析方面表现优异。相比之下, 矩形特征虽然对边缘、条形结构及其他简单图像结构的存在较为敏感, 但其表征能力相对粗糙。与可操纵滤波器不同, 矩形特征仅支持垂直、水平和对角线方向。然而, 矩形特征集合确实能提供丰富的图像表征, 足以支撑有效的学习过程。结合积分图像技术, 矩形特征集的高效性充分弥补了其灵活性不足的缺陷。

3. 学习分类函数

给定一个特征集和一个包含正负图像样本的训练集, 可以采用多种机器学习方法

可用于学习分类函数。在我们的系统中, 采用了一种AdaBoost的变体both来选择一小部分特征并训练分类器[6]。最初形式的AdaBoost学习算法旨在提升简单(有时称为弱)学习算法的分类性能。AdaBoost学习过程提供了若干形式化保证。Freund和Schapire证明了强分类器的训练误差随着迭代轮数呈指数级趋近于零。更重要的是, 后续研究证明了关于泛化性能的若干结果[14]。其关键见解在于, 泛化性能与样本间隔相关, 而AdaBoost能够快速实现较大的间隔。

回想一下, 每个图像子窗口关联着超过18万个矩形特征, 这个数字远大于像素数量。尽管每个特征的计算效率很高, 但计算完整特征集的成本却高得令人望而却步。我们的假设(已通过实验验证)是, 只需极少量这类特征组合便能构成一个有效的分类器。主要挑战在于如何找出这些特征。

为支持这一目标, 弱学习算法被设计用于选择最能区分正负样本的单个矩形特征(这与[2]在图像数据库检索领域采用的方法类似)。对于每个特征, 弱学习器会确定最优阈值分类函数, 以使误分类的样本数量最小化。因此, 一个弱分类器 $f_i(x)$ 由特征 μ_i , 阈值 θ_i 以及指示不等式方向的奇偶性 b_i 组成:

$$f_i(x) = \begin{cases} 1 & \text{if } b_i \mu_i(x) < b_i \theta_i \\ 0 & \text{otherwise} \end{cases}$$

这里 $\mathcal{I}$ 是图像的一个24x24像素子窗口。关于提升过程的总结, 请参见表1。

在实践中, 没有单一特征能以低误差完成分类任务。在提升过程的早期轮次中被选中的特征, 其错误率介于0.1至0.3之间。随着任务难度增加, 在后续轮次中被选中的特征则会产生0.4至0.5的错误率。

3.1. 学习讨论

许多通用的特征选择方法已被提出(综述见[18]第8章)。我们的最终应用需要一种极为激进的方法, 以舍弃绝大多数特征。针对类似的识别问题, Papageorgiou等人提出了一种基于特征方差的特征选择方案[10]。他们展示了从总计1734个特征中仅选择37个特征仍能取得良好效果。

Roth等人提出了一种基于Winnow指数感知器学习规则的特征选择过程[11]。Winnow学习过程收敛于一个解, 其中许多权重为零。然而, 一个非常庞大的

- Given example images $(x_1, y_1), \dots, (x_n, y_n)$ where $y_i = 0, 1$ for negative and positive examples respectively.
- Initialize weights $w_{1,i} = \frac{1}{2m}, \frac{1}{2l}$ for $y_i = 0, 1$ respectively, where m and l are the number of negatives and positives respectively.
- For $t = 1, \dots, T$:

1. Normalize the weights,

$$w_{t,i} \leftarrow \frac{w_{t,i}}{\sum_{j=1}^n w_{t,j}}$$

so that w_t is a probability distribution.

2. For each feature, j , train a classifier h_j which is restricted to using a single feature. The error is evaluated with respect to w_t , $\epsilon_j = \sum_i w_i |h_j(x_i) - y_i|$.
3. Choose the classifier, h_t , with the lowest error ϵ_t .
4. Update the weights:

$$w_{t+1,i} = w_{t,i} \beta_t^{1-e_i}$$

where $e_i = 0$ if example x_i is classified correctly, $e_i = 1$ otherwise, and $\beta_t = \frac{\epsilon_t}{1-\epsilon_t}$.

- The final strong classifier is:

$$h(x) = \begin{cases} 1 & \sum_{t=1}^T \alpha_t h_t(x) \geq \frac{1}{2} \sum_{t=1}^T \alpha_t \\ 0 & \text{otherwise} \end{cases}$$

where $\alpha_t = \log \frac{1}{\beta_t}$

Table 1: The AdaBoost algorithm for classifier learning. Each round of boosting selects one feature from the 180,000 potential features.

number of features are retained (perhaps a few hundred or thousand).

3.2. Learning Results

While details on the training and performance of the final system are presented in Section 5, several simple results merit discussion. Initial experiments demonstrated that a frontal face classifier constructed from 200 features yields a detection rate of 95% with a false positive rate of 1 in 14084. These results are compelling, but not sufficient for many real-world tasks. In terms of computation, this classifier is probably faster than any other published system, requiring 0.7 seconds to scan an 384 by 288 pixel image. Unfortunately, the most straightforward technique for improving detection performance, adding features to the classifier, directly increases computation time.

For the task of face detection, the initial rectangle features selected by AdaBoost are meaningful and easily interpreted. The first feature selected seems to focus on the property that the region of the eyes is often darker than the region

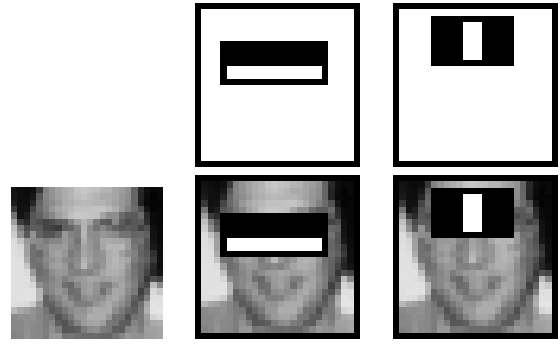


Figure 3: The first and second features selected by AdaBoost. The two features are shown in the top row and then overlayed on a typical training face in the bottom row. The first feature measures the difference in intensity between the region of the eyes and a region across the upper cheeks. The feature capitalizes on the observation that the eye region is often darker than the cheeks. The second feature compares the intensities in the eye regions to the intensity across the bridge of the nose.

of the nose and cheeks (see Figure 3). This feature is relatively large in comparison with the detection sub-window, and should be somewhat insensitive to size and location of the face. The second feature selected relies on the property that the eyes are darker than the bridge of the nose.

4. The Attentional Cascade

This section describes an algorithm for constructing a cascade of classifiers which achieves increased detection performance while radically reducing computation time. The key insight is that smaller, and therefore more efficient, boosted classifiers can be constructed which reject many of the negative sub-windows while detecting almost all positive instances (i.e. the threshold of a boosted classifier can be adjusted so that the false negative rate is close to zero). Simpler classifiers are used to reject the majority of sub-windows before more complex classifiers are called upon to achieve low false positive rates.

The overall form of the detection process is that of a degenerate decision tree, what we call a “cascade” (see Figure 4). A positive result from the first classifier triggers the evaluation of a second classifier which has also been adjusted to achieve very high detection rates. A positive result from the second classifier triggers a third classifier, and so on. A negative outcome at any point leads to the immediate rejection of the sub-window.

Stages in the cascade are constructed by training classifiers using AdaBoost and then adjusting the threshold to minimize false negatives. Note that the default AdaBoost threshold is designed to yield a low error rate on the training data. In general a lower threshold yields higher detec-

- Given example images $(x^1, y^1), \dots, (x^N, y^N)$ where $y^i \in \{-1, 1\}$ for negative and positive examples respectively. Initialize weights $w_i = 1/N$ for $i = 1, \dots, N$ respectively, where N and N are the number of negatives and positives respectively.
- For $t = 1, 2, \dots$:
 1. Normalize the weights,

$$w_i^t \leftarrow \frac{w_i^{t-1}}{\sum_{j=1}^N w_j^{t-1}}$$
 so that w_i^t is a probability distribution.
 2. For each feature, f , train a classifier h_f which is restricted to using a single feature. The error is evaluated with respect to w_i^t ,

$$\epsilon_f = \sum_{i=1}^N w_i^t \mathbb{1}_{h_f(x^i) \neq y^i}$$
 3. Choose the classifier, h_{f^*} , with the lowest error ϵ_{f^*} .
 4. Update the weights:

$$w_i^{t+1} = w_i^t \exp(-\epsilon_f^t y^i)$$
 where $\epsilon_f^t = \epsilon_{f^*}$ if example i is classified correctly, otherwise, $\epsilon_f^t = \epsilon_{f^*} + 1$.
- The final strong classifier is:

$$H(x) = \begin{cases} 1 & \sum_{f=1}^F \alpha_f h_f(x) > \frac{1}{2} \sum_{f=1}^F \alpha_f \\ 0 & \text{otherwise} \end{cases}$$
 where $\alpha_f = \log \frac{1}{\epsilon_f}$

表1: AdaBoost分类器学习算法。每一轮提升从180,000个潜在特征中选择一个特征。

保留的特征数量（可能为几百或几千个）。

3.2. 学习结果

尽管最终系统的训练和性能细节将在第5节中介绍，但几项简单结果值得探讨。初步实验表明，由200个特征构建的正面人脸分类器可实现95%的检测率，每14084次检测中出现1次误报。这些结果令人瞩目，但对于许多实际任务仍显不足。在计算效率方面，该分类器可能比任何已公开的系统更快速，扫描一张384×288像素的图像仅需0.7秒。然而，提升检测性能最直接的方法——为分类器增加特征——会直接导致计算时间增加。

在人脸检测任务中，AdaBoost算法选取的初始矩形特征具有明确意义且易于解读。首个被选中的特征似乎聚焦于眼睛区域通常比脸颊区域更暗的特性。

图3: AdaBoost选择的第一和第二个特征。这两个特征在顶行显示，然后在底行叠加在一个典型的训练人脸图像上。第一个特征测量眼睛区域与上脸颊区域之间的强度差异。该特征利用了眼睛区域通常比脸颊更暗的观察结果。第二个特征比较眼睛区域的强度与鼻梁区域的强度。

鼻子和脸颊（见图3）。这一特征相对于检测子窗口来说比较大，并且应该对脸部的大小和位置有一定的鲁棒性。第二个选择的特征利用了眼睛比鼻梁更暗的特性。

4. 注意力级联

本节描述了一种构建级联分类器的算法，该算法在显著减少计算时间的同时提升了检测性能。其核心思路是：可以构建更小、因而更高效的增强分类器，在检测到几乎所有正例样本（即通过调整增强分类器的阈值使漏检率接近零）的同时，能够拒绝大量负子窗口。该算法先使用较简单的分类器排除大部分子窗口，再调用更复杂的分类器来实现较低误报率。

检测过程的整体形式是一种退化的决策树，我们称之为“级联”（见图4）。第一个分类器的肯定结果会触发第二个分类器的评估，该分类器同样经过调整以实现极高的检测率。第二个分类器的肯定结果会触发第三个分类器，依此类推。在任何阶段出现否定结果都会导致子窗口被立即拒绝。

级联检测的各个阶段通过使用AdaBoost训练分类器构建，随后调整阈值以最小化漏检率。需要注意的是，默认的AdaBoost阈值旨在使训练数据上的错误率保持较低水平。一般而言，较低的阈值会带来更高的检测——

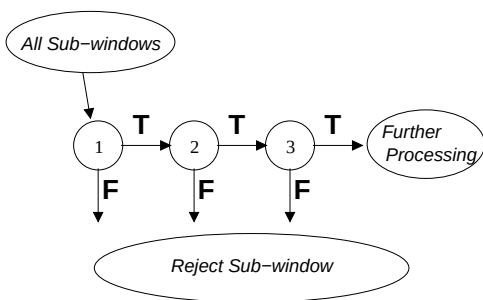


Figure 4: Schematic depiction of a the detection cascade. A series of classifiers are applied to every sub-window. The initial classifier eliminates a large number of negative examples with very little processing. Subsequent layers eliminate additional negatives but require additional computation. After several stages of processing the number of sub-windows have been reduced radically. Further processing can take any form such as additional stages of the cascade (as in our detection system) or an alternative detection system.

tion rates and higher false positive rates.

For example an excellent first stage classifier can be constructed from a two-feature strong classifier by reducing the threshold to minimize false negatives. Measured against a validation training set, the threshold can be adjusted to detect 100% of the faces with a false positive rate of 40%. See Figure 3 for a description of the two features used in this classifier.

Computation of the two feature classifier amounts to about 60 microprocessor instructions. It seems hard to imagine that any simpler filter could achieve higher rejection rates. By comparison, scanning a simple image template, or a single layer perceptron, would require at least 20 times as many operations per sub-window.

The structure of the cascade reflects the fact that within any single image an overwhelming majority of sub-windows are negative. As such, the cascade attempts to reject as many negatives as possible at the earliest stage possible. While a positive instance will trigger the evaluation of every classifier in the cascade, this is an exceedingly rare event.

Much like a decision tree, subsequent classifiers are trained using those examples which pass through all the previous stages. As a result, the second classifier faces a more difficult task than the first. The examples which make it through the first stage are “harder” than typical examples. The more difficult examples faced by deeper classifiers push the entire receiver operating characteristic (ROC) curve downward. At a given detection rate, deeper classifiers have correspondingly higher false positive rates.

4.1. Training a Cascade of Classifiers

The cascade training process involves two types of trade-offs. In most cases classifiers with more features will achieve higher detection rates and lower false positive rates. At the same time classifiers with more features require more time to compute. In principle one could define an optimization framework in which: i) the number of classifier stages, ii) the number of features in each stage, and iii) the threshold of each stage, are traded off in order to minimize the expected number of evaluated features. Unfortunately finding this optimum is a tremendously difficult problem.

In practice a very simple framework is used to produce an effective classifier which is highly efficient. Each stage in the cascade reduces the false positive rate and decreases the detection rate. A target is selected for the minimum reduction in false positives and the maximum decrease in detection. Each stage is trained by adding features until the target detection and false positives rates are met (these rates are determined by testing the detector on a validation set). Stages are added until the overall target for false positive and detection rate is met.

4.2. Detector Cascade Discussion

The complete face detection cascade has 38 stages with over 6000 features. Nevertheless the cascade structure results in fast average detection times. On a difficult dataset, containing 507 faces and 75 million sub-windows, faces are detected using an average of 10 feature evaluations per sub-window. In comparison, this system is about 15 times faster than an implementation of the detection system constructed by Rowley et al.³ [12]

A notion similar to the cascade appears in the face detection system described by Rowley et al. in which two detection networks are used [12]. Rowley et al. used a faster yet less accurate network to prescreen the image in order to find candidate regions for a slower more accurate network. Though it is difficult to determine exactly, it appears that Rowley et al.’s two network face system is the fastest existing face detector.⁴

The structure of the cascaded detection process is essentially that of a degenerate decision tree, and as such is related to the work of Amit and Geman [1]. Unlike techniques which use a fixed detector, Amit and Geman propose an alternative point of view where unusual co-occurrences of simple image features are used to trigger the evaluation of a more complex detection process. In this way the full detection process need not be evaluated at many of the potential image locations and scales. While this basic insight

³Henry Rowley very graciously supplied us with implementations of his detection system for direct comparison. Reported results are against his fastest system. It is difficult to determine from the published literature, but the Rowley-Baluja-Kanade detector is widely considered the fastest detection system and has been heavily tested on real-world problems.

⁴Other published detectors have either neglected to discuss performance in detail, or have never published detection and false positive rates on a large and difficult training set.

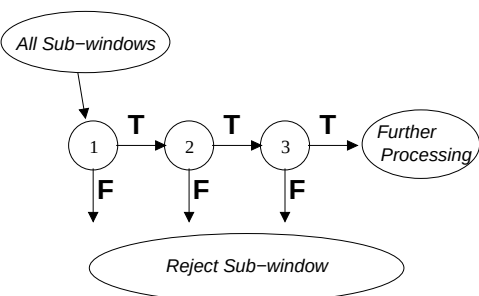


图4：检测级联的示意图。一系列分类器被应用于每个子窗口。初始分类器以极少的处理量消除了大量负样本。后续层级消除了更多负样本但需要额外计算。经过若干处理阶段后，子窗口数量已急剧减少。后续处理可采用任何形式，例如级联的附加阶段（如我们的检测系统所示）或替代检测系统。

检出率较低且假阳性率较高。

例如，一个优秀的第一级分类器可以通过降低阈值来最小化假阴性，从而由双特征强分类器构建。在验证训练集上测量时，可以调整阈值以检测100%的人脸，同时假阳性率为40%。有关此分类器中使用的两个特征的描述，请参见图3。

两个特征分类器的计算量大约相当于60条微处理器指令。很难想象还有更简单的滤波器能达到更高的拒绝率。相比之下，扫描一个简单的图像模板或单层感知器，每个子窗口至少需要20倍以上的操作量。

级联结构反映了这样一个事实：在任意单张图像中，绝大多数子窗口都是负样本。因此，级联系统试图在最早阶段尽可能多地排除负样本。虽然正样本会触发级联中每个分类器的评估，但这种情况极为罕见。

与决策树类似，后续分类器使用那些通过所有先前阶段的样本进行训练。因此，第二个分类器面临的任务比第一个更困难。通过第一阶段的样本比典型样本“更难”。更深层分类器所面临的更困难样本将整个接收者操作特征（ROC）曲线向下推移。在给定的检测率下，更深层的分类器相应地具有更高的误报率。

4.1. 训练级联分类器

级联训练过程涉及两种权衡。在大多数情况下，具有更多特征的分类器会获得更高的检测率和更低的误报率。同时，具有更多特征的分类器需要更多的计算时间。原则上，可以定义一个优化框架，其中：i) 分类器阶段的数量，ii) 每个阶段的特征数量，以及 iii) 每个阶段的阈值，被用来权衡以最小化预期评估特征的数量。不幸的是，找到这个最优解是一个极其困难的问题。

在实践中，采用一个非常简单的框架来生成高效且效果显著的分类器。级联中的每个阶段都会降低误报率并减少检测率。选择目标时，需追求误报率的最小化降低和检测率的最大化减少。每个阶段通过不断添加特征进行训练，直至达到目标检测率和误报率（这些比率通过在验证集上测试检测器来确定）。持续添加阶段，直至满足整体误报率和检测率的目标。

4.2. 检测器级联讨论

完整的人脸检测级联包含38个阶段，超过6000个特征。尽管如此，级联结构仍能实现快速的检测平均时间。在一个包含507张人脸和7500万个子窗口的困难数据集上，每个子窗口平均仅需10次特征评估即可检测出人脸。相比之下，该系统比Rowley等人³[12]构建的检测系统实现快约15倍。

一种类似于级联的概念出现在Rowley等人描述的人脸检测系统中，该系统使用了两个检测网络[12]。Rowley等人使用了一个更快但准确性较低的网络对图像进行预筛选，以便为更慢、更准确的网络找到候选区域。尽管难以精确确定，但Rowley等人的双网络人脸系统似乎是现有最快的人脸检测器。{v*}

级联检测过程的结构本质上是一个退化决策树，因此与Amit和Geman的研究[1]相关。与使用固定检测器的技术不同，Amit和Geman提出了一种替代视角，即利用简单图像特征的异常共现来触发更复杂检测过程的评估。通过这种方式，无需在大量潜在图像位置和尺度上评估完整的检测过程。虽然这一基本见解

³Henry Rowley very graciously supplied us with implementations of his detection system for direct comparison. Reported results are against his fastest system. It is difficult to determine from the published literature, but the Rowley-Baluja-Kanade detector is widely considered the fastest detection system and has been heavily tested on real-world problems.

⁴Other published detectors have either neglected to discuss performance in detail, or have never published detection and false positive rates on a large and difficult training set.

is very valuable, in their implementation it is necessary to first evaluate some feature detector at every location. These features are then grouped to find unusual co-occurrences. In practice, since the form of our detector and the features that it uses are extremely efficient, the amortized cost of evaluating our detector at *every scale and location* is much faster than finding and grouping edges throughout the image.

In recent work Fleuret and Geman have presented a face detection technique which relies on a “chain” of tests in order to signify the presence of a face at a particular scale and location [4]. The image properties measured by Fleuret and Geman, disjunctions of fine scale edges, are quite different than rectangle features which are simple, exist at all scales, and are somewhat interpretable. The two approaches also differ radically in their learning philosophy. The motivation for Fleuret and Geman’s learning process is density estimation and density discrimination, while our detector is purely discriminative. Finally the false positive rate of Fleuret and Geman’s approach appears to be higher than that of previous approaches like Rowley et al. and this approach. Unfortunately the paper does not report quantitative results of this kind. The included example images each have between 2 and 10 false positives.

5 Results

A 38 layer cascaded classifier was trained to detect frontal upright faces. To train the detector, a set of face and non-face training images were used. The face training set consisted of 4916 hand labeled faces scaled and aligned to a base resolution of 24 by 24 pixels. The faces were extracted from images downloaded during a random crawl of the world wide web. Some typical face examples are shown in Figure 5. The non-face subwindows used to train the detector come from 9544 images which were manually inspected and found to not contain any faces. There are about 350 million subwindows within these non-face images.

The number of features in the first five layers of the detector is 1, 10, 25, 25 and 50 features respectively. The remaining layers have increasingly more features. The total number of features in all layers is 6061.

Each classifier in the cascade was trained with the 4916 training faces (plus their vertical mirror images for a total of 9832 training faces) and 10,000 non-face sub-windows (also of size 24 by 24 pixels) using the Adaboost training procedure. For the initial one feature classifier, the non-face training examples were collected by selecting random sub-windows from a set of 9544 images which did not contain faces. The non-face examples used to train subsequent layers were obtained by scanning the partial cascade across the non-face images and collecting false positives. A maximum of 10000 such non-face sub-windows were collected for each layer.

Speed of the Final Detector



Figure 5: Example of frontal upright face images used for training.

The speed of the cascaded detector is directly related to the number of features evaluated per scanned sub-window. Evaluated on the MIT+CMU test set [12], an average of 10 features out of a total of 6061 are evaluated per sub-window. This is possible because a large majority of sub-windows are rejected by the first or second layer in the cascade. On a 700 Mhz Pentium III processor, the face detector can process a 384 by 288 pixel image in about .067 seconds (using a starting scale of 1.25 and a step size of 1.5 described below). This is roughly 15 times faster than the Rowley-Baluja-Kanade detector [12] and about 600 times faster than the Schneiderman-Kanade detector [15].

Image Processing

All example sub-windows used for training were variance normalized to minimize the effect of different lighting conditions. Normalization is therefore necessary during detection as well. The variance of an image sub-window can be computed quickly using a pair of integral images. Recall that $\sigma^2 = m^2 - \frac{1}{N} \sum x^2$, where σ is the standard deviation, m is the mean, and x is the pixel value within the sub-window. The mean of a sub-window can be computed using the integral image. The sum of squared pixels is computed using an integral image of the image squared (i.e. two integral images are used in the scanning process). During scanning the effect of image normalization can be achieved by post-multiplying the feature values rather than pre-multiplying the pixels.

Scanning the Detector

The final detector is scanned across the image at multiple scales and locations. Scaling is achieved by scaling the detector itself, rather than scaling the image. This process makes sense because the features can be evaluated at any

非常有价值，在实施过程中，首先需要在每个位置评估某种特征检测器。然后对这些特征进行分组，以发现不寻常的共现现象。实际上，由于我们的检测器形式及其使用的特征极其高效，在*every scale and location*处评估检测器的均摊成本远低于在整个图像中寻找并分组边缘。

在最近的研究中，Fleuret和Geman提出了一种人脸检测技术，该技术依赖于一系列测试的“链”来指示特定尺度和位置是否存在人脸[4]。Fleuret和Geman所测量的图像属性——精细尺度边缘的析取——与矩形特征截然不同，后者简单、存在于所有尺度，且具有一定可解释性。这两种方法在学习理念上也存在根本差异。Fleuret和Geman学习过程的动机是密度估计和密度判别，而我们的检测器纯粹是判别式的。最后，Fleuret和Geman方法的误报率似乎高于以往的方法，如Rowley等人的方法以及本方法。遗憾的是，该论文未报告此类定量结果。所包含的示例图像每张都有2到10个误报。

5 结果

一个38层的级联分类器被训练用于检测正面直立的人脸。为了训练这个检测器，使用了一组人脸和非人脸的训练图像。人脸训练集包含4916张手动标记的人脸图像，这些图像被缩放并对齐到24×24像素的基础分辨率。这些人脸图像是从随机爬取互联网时下载的图片中提取的。图5展示了一些典型的人脸示例。用于训练检测器的非人脸子窗口来自9544张经过人工检查确认不包含任何人脸的图像。这些非人脸图像中大约有3.5亿个子窗口。

检测器前五层的特征数量分别为1、10、25、25和50个特征。其余各层具有越来越多的特征。所有层中的特征总数为6061个。

级联中的每个分类器均使用Adaboost训练流程，基于4916张训练人脸（加上其垂直镜像图像，总计9832张训练人脸）和10000个非人脸子窗口（同样为24×24像素大小）进行训练。对于初始的单特征分类器，非人脸训练样本通过从一组不含人脸的9544张图像中随机选取子窗口来收集。用于训练后续层的非人脸样本，则是通过将部分级联分类器在非人脸图像上进行扫描并收集误报结果而获得。每层最多收集10000个此类非人脸子窗口。

最终检测器的速度



图5：用于训练的正面直立人脸图像示例。

级联检测器的速度直接与每个扫描子窗口评估的特征数量相关。在MIT+CMU测试集[12]上的评估显示，每个子窗口平均评估6061个特征中的10个。这是可能的，因为绝大多数子窗口在级联的第一层或第二层就被拒绝了。在700 MHz的Pentium III处理器上，人脸检测器处理一张384×288像素的图像大约需要0.067秒（使用起始尺度为1.25、步长为1.5的参数设置，详见下文）。这比Rowley-Baluja-Kanade检测器[12]快了约15倍，比Schneiderman-Kanade检测器[15]快了约600倍。

图像处理

所有用于训练的示例子窗口都经过方差归一化处理，以最小化不同光照条件的影响。因此，在检测过程中同样需要进行归一化。图像子窗口的方差可以借助一对积分图像快速计算。回顾公式 $\sigma_q^2 = w_q^2 - \frac{1}{A} \sum x_q^2$ ，其中 σ_q 为标准差， w_q 为均值， x_q 为子窗口内的像素值。子窗口的均值可通过积分图像计算，而像素平方和则通过图像平方的积分图像（即在扫描过程中使用两个积分图像）计算。在扫描过程中，图像归一化的效果可以通过后乘特征值而非预乘像素值来实现。

扫描探测器

最终的检测器在多个尺度和位置上对图像进行扫描。缩放是通过缩放检测器本身实现的，而非缩放图像。这一过程是合理的，因为特征可以在任意

False detections Detector	10	31	50	65	78	95	167
Viola-Jones	76.1%	88.4%	91.4%	92.0%	92.1%	92.9%	93.9%
Viola-Jones (voting)	81.1%	89.7%	92.1%	93.1%	93.1%	93.2%	93.7%
Rowley-Baluja-Kanade	83.2%	86.0%	-	-	-	89.2%	90.1%
Schneiderman-Kanade	-	-	-	94.4%	-	-	-
Roth-Yang-Ahuja	-	-	-	-	(94.8%)	-	-

Table 2: Detection rates for various numbers of false positives on the MIT+CMU test set containing 130 images and 507 faces.

scale with the same cost. Good results were obtained using a set of scales a factor of 1.25 apart.

The detector is also scanned across location. Subsequent locations are obtained by shifting the window some number of pixels Δ . This shifting process is affected by the scale of the detector: if the current scale is s the window is shifted by $\lceil s\Delta \rceil$, where $\lceil \cdot \rceil$ is the rounding operation.

The choice of Δ affects both the speed of the detector as well as accuracy. The results we present are for $\Delta = 1.0$. We can achieve a significant speedup by setting $\Delta = 1.5$ with only a slight decrease in accuracy.

Integration of Multiple Detections

Since the final detector is insensitive to small changes in translation and scale, multiple detections will usually occur around each face in a scanned image. The same is often true of some types of false positives. In practice it often makes sense to return one final detection per face. Toward this end it is useful to postprocess the detected sub-windows in order to combine overlapping detections into a single detection.

In these experiments detections are combined in a very simple fashion. The set of detections are first partitioned into disjoint subsets. Two detections are in the same subset if their bounding regions overlap. Each partition yields a single final detection. The corners of the final bounding region are the average of the corners of all detections in the set.

Experiments on a Real-World Test Set

We tested our system on the MIT+CMU frontal face test set [12]. This set consists of 130 images with 507 labeled frontal faces. A ROC curve showing the performance of our detector on this test set is shown in Figure 6. To create the ROC curve the threshold of the final layer classifier is adjusted from $-\infty$ to $+\infty$. Adjusting the threshold to $+\infty$ will yield a detection rate of 0.0 and a false positive rate of 0.0. Adjusting the threshold to $-\infty$, however, increases both the detection rate and false positive rate, but only to a certain point. Neither rate can be higher than the rate of the detection cascade minus the final layer. In effect, a threshold of $-\infty$ is equivalent to removing that layer. Further increasing the detection and false positive rates requires decreasing the threshold of the next classifier in the cascade.

Thus, in order to construct a complete ROC curve, classifier layers are removed. We use the *number* of false positives as opposed to the *rate* of false positives for the x-axis of the ROC curve to facilitate comparison with other systems. To compute the false positive rate, simply divide by the total number of sub-windows scanned. In our experiments, the number of sub-windows scanned is 75,081,800.

Unfortunately, most previous published results on face detection have only included a single operating regime (i.e. single point on the ROC curve). To make comparison with our detector easier we have listed our detection rate for the false positive rates reported by the other systems. Table 2 lists the detection rate for various numbers of false detections for our system as well as other published systems. For the Rowley-Baluja-Kanade results [12], a number of different versions of their detector were tested yielding a number of different results they are all listed in under the same heading. For the Roth-Yang-Ahuja detector [11], they reported their result on the MIT+CMU test set minus 5 images containing line drawn faces removed.

Figure 7 shows the output of our face detector on some test images from the MIT+CMU test set.

A simple voting scheme to further improve results

In table 2 we also show results from running three detectors (the 38 layer one described above plus two similarly trained detectors) and outputting the majority vote of the three detectors. This improves the detection rate as well as eliminating more false positives. The improvement would be greater if the detectors were more independent. The correlation of their errors results in a modest improvement over the best single detector.

6 Conclusions

We have presented an approach for object detection which minimizes computation time while achieving high detection accuracy. The approach was used to construct a face detection system which is approximately 15 faster than any previous approach.

This paper brings together new algorithms, representations, and insights which are quite generic and may well

Detector	False detections						
	10	31	50	65	78	95	167
Viola-Jones	76.1%	88.4%	91.4%	92.0%	92.1%	92.9%	93.9%
Viola-Jones (voting)	81.1%	89.7%	92.1%	93.1%	93.1%	93.2%	93.7%
Rowley-Baluja-Kanade	83.2%	86.0%	-	-	-	89.2%	90.1%
Schneiderman-Kanade	-	-	-	94.4%	-	-	-
Roth-Yang-Ahuja	-	-	-	-	(94.8%)	-	-

表2：在包含130张图像和50个的MIT+CMU测试集上，不同误报数量对应的检测率

7

以相同的成本进行缩放。使用一组间隔1.25倍的缩放比例获得了良好的结果。

检测器也会在位置上进行扫描。后续位置通过将窗口移动若干像素来获得。这一移动过程受检测器尺度的影响：若当前尺度为 s ，则窗口移动步长为 $\lceil \frac{s}{4} \rceil$ ，其中 $\lceil \cdot \rceil$ 表示取整运算。

的选择会影响检测器的速度和准确性。我们展示的结果是针对 $s = 1$ 的。通过设置 $s = 1.25$ ，我们可以在精度仅有轻微下降的情况下实现显著的速度提升。

多检测集成

由于最终检测器对平移和尺度的小变化不敏感，因此在扫描图像中，每张人脸周围通常会出现多个检测结果。对于某些类型的误报，情况也往往如此。实际上，通常更合理的做法是每张人脸返回一个最终检测结果。为此，对检测到的子窗口进行后处理是有益的，以便将重叠的检测合并为单一检测。

在这些实验中，检测结果以非常简单的方式合并。首先将检测集合划分为互不相交的子集。若两个检测的边界区域重叠，则它们属于同一子集。每个子集生成一个最终检测结果，最终边界区域的角点取该子集中所有检测角点的平均值。

真实世界测试集上的实验

我们在MIT+CMU正面人脸测试集[12]上评估了我们的系统。该测试集包含130张图像，共标注了507个正面人脸。图6展示了我们的检测器在此测试集上的ROC性能曲线。为生成ROC曲线，我们将最终层分类器的阈值从 $-\infty$ 调整至 $+\infty$ 。将阈值设为 $-\infty$ 时，检测率与误报率均为0.0；而将阈值调整至 $+\infty$ 则会使检测率和误报率同时上升，但仅能提升至特定上限——两者均无法超过检测级联中去除最终层后的基准水平。实际上，将阈值设为 $-\infty$ 等效于移除该层级。若要进一步提升检测率与误报率，则需降低级联中下一级分类器的阈值。

因此，为了构建完整的ROC曲线，分类器层被移除。

我们使用假阳性的 $number$ 而非假阳性的 $rate$ 作为ROC曲线的x轴，以便与其他系统进行比较。计算假阳性率时，只需将其除以扫描的子窗口总数。在我们的实验中，扫描的子窗口数量为75,081,800个。

遗憾的是，以往发表的人脸检测结果大多只包含单一操作区间（即ROC曲线上的单个点）。为便于与我们的检测器进行比较，我们列出了其他系统所报告的误报率对应的检测率。表2列出了我们的系统以及其它已发表系统在不同误检数量下的检测率。对于Rowley-Baluja-Kanade的研究成果[12]，其检测器的多个不同版本经过测试产生了若干差异结果，这些数据均列于同一标题下。至于Roth-Yang-Ahuja检测器[11]，其报告结果基于MIT+CMU测试集（该数据集已移除5幅含线条绘制人脸的图像）。

图7展示了我们的人脸检测器在MIT+CMU测试集部分测试图像上的输出结果。

一种简单的投票方案，以进一步提升结果

在表2中，我们还展示了运行三个检测器（上文描述的38层检测器以及两个经过类似训练的检测器）并输出三者多数投票的结果。这不仅提高了检测率，还减少了更多误报。如果这些检测器更具独立性，改进效果会更显著。由于它们的误差存在相关性，当前仅比最佳单检测器取得了有限的改进。

6 结论

我们提出了一种目标检测方法，该方法在实现高检测精度的同时最大限度地减少了计算时间。该方法被用于构建一个人脸检测系统，其速度比以往任何方法快约15倍。

本文汇集了新的算法、表示方法和见解，这些内容具有相当的通用性，很可能

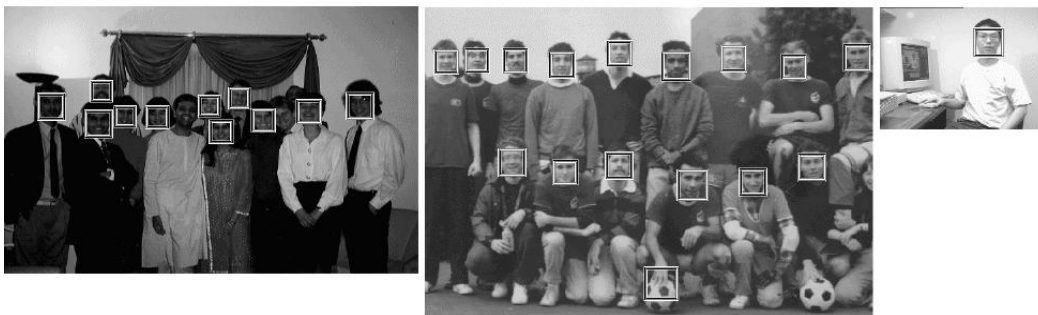


Figure 7: Output of our face detector on a number of test images from the MIT+CMU test set.

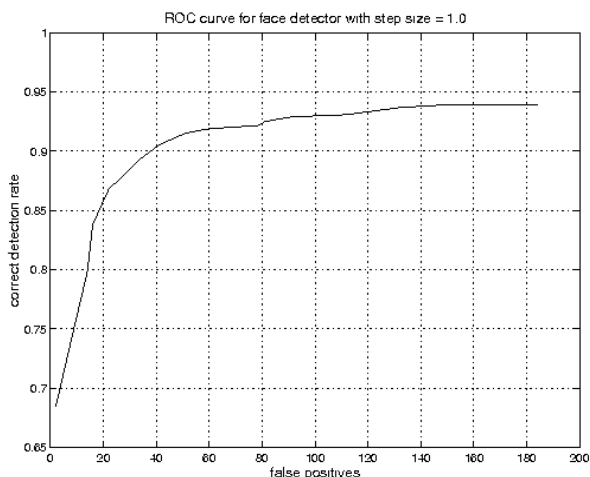


Figure 6: ROC curve for our face detector on the MIT+CMU test set. The detector was run using a step size of 1.0 and starting scale of 1.0 (75,081,800 sub-windows scanned).

have broader application in computer vision and image processing.

Finally this paper presents a set of detailed experiments on a difficult face detection dataset which has been widely studied. This dataset includes faces under a very wide range of conditions including: illumination, scale, pose, and camera variation. Experiments on such a large and complex dataset are difficult and time consuming. Nevertheless systems which work under these conditions are unlikely to be brittle or limited to a single set of conditions. More importantly conclusions drawn from this dataset are unlikely to be experimental artifacts.

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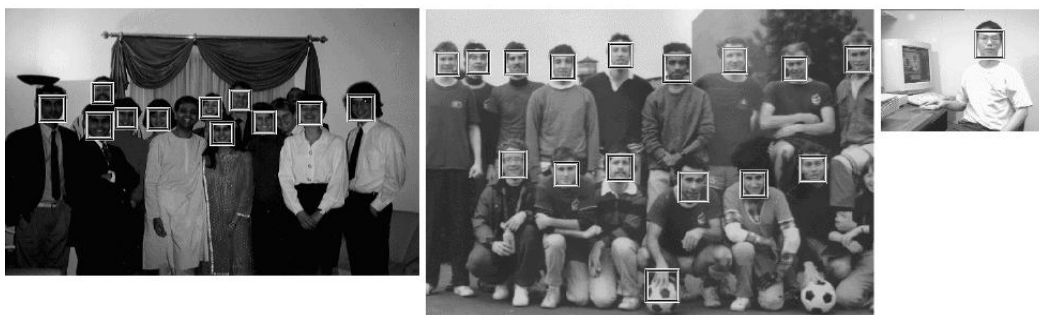


图7：我们的面部检测器在MIT+CMU测试集若干测试图像上的输出结果。

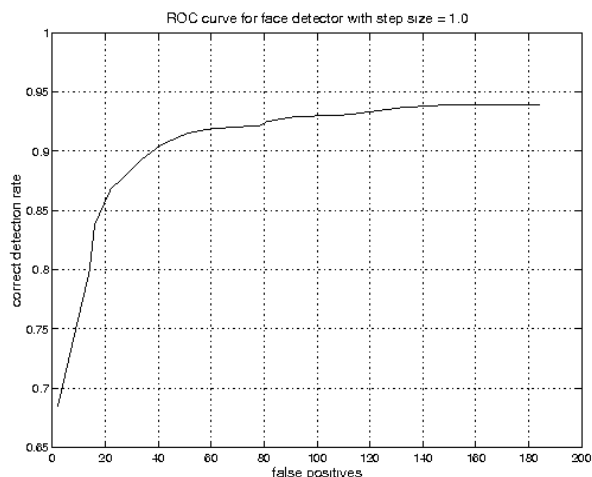


图6：我们的人脸检测器在MIT+CMU测试集上的ROC曲线。检测器运行时采用1.0的步长和1.0的起始尺度（共扫描75,081,800个子窗口）。

在计算机视觉和图像处理领域有更广泛的应用。

最后，本文在一个已被广泛研究的困难人脸检测数据集上进行了一系列详细实验。该数据集涵盖了极广泛条件下的人脸图像，包括：光照、尺度、姿态及相机差异。在此类庞大而复杂的数据集上进行实验既困难又耗时。然而，能在这些条件下运行的系统不太可能脆弱或仅局限于单一条件集。更重要的是，基于该数据集得出的结论不太可能是实验假象。

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